

Senolytic Selectivity Is a Differential-Dependency Quantity, Not a Binding Affinity

A Closed-Form Selectivity Law, a Heterogeneity Ceiling Theorem, and an Orthogonal Three-Axis AND-Gate Design (SENOLYX-AG) for Regenerative-Niche Senescent Fibroblasts

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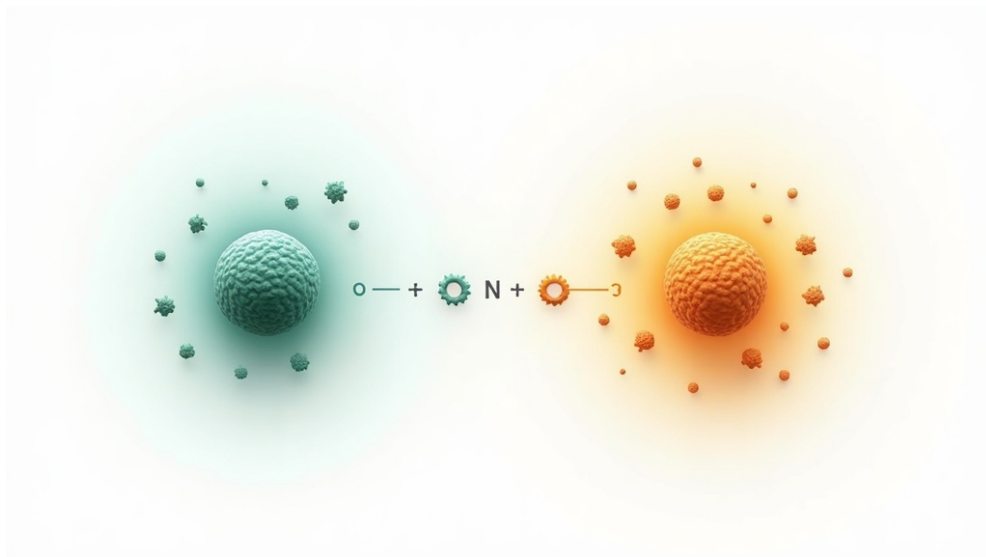


Figure 1: The SENOLYX-AG concept (fal.ai/FLUX-generated cover). A healthy cell (left) engages at most one recognition lock and is spared; a senescent cell (right) engages all three orthogonal locks (two surface, one lysosomal) and is cleared. Selectivity is the multiplicative AND of independent recognition axes.

Abstract

We report a closed-form theory of *senolytic selectivity* — the therapeutic window separating senescent-cell killing from healthy-cell toxicity — together with a concrete, in-silico-supported molecular design that follows from it. The work was prompted by a campaign wall: an absolute binding free-energy (ABFE) effort to optimize senolytic candidates plateaued at the force-field ceiling. We show this was not a numerical failure but a *category error*. Two independent “wrong axes” are identified and ruled out. First, selectivity is governed by the *differential anti-apoptotic dependency* between senescent and quiescent cells, f_s/f_q , and the absolute binding free energy ΔG_{bind} *cancels* from the selectivity condition — explaining mechanistically why the ABFE axis was uninformative, and corroborated by the textbook navitoclax counterexample (tightest BCL- x_L affinity, narrowest clinical window). Second, the downstream “% senescent cells cleared $\rightarrow$ regeneration” transfer function is non-monotone and not literature-grounded; clearance-% is itself a poor gate variable, replaced

here by a functional η_{neo} pharmacodynamic model anchored to two real clinical outcomes (UBX1325 success, UBX0101 failure). We derive a selectivity law $x^* = (B - A)/(f B)$ with a selectivity condition $\Delta x^* > 0$; a heterogeneity *ceiling theorem* $f_{\text{clear}}(T) \leq p_{\text{dep}}(T) < 1$ for any single-target agent (no universal senescence dependency exists); and a multiplicative *escape* $S_{\text{total}} = \prod_i S_i$ available only on orthogonal recognition axes. A direct numerical evaluation (the **S1** index) shows a single marker is net anti-selective ($0.92\times$), a two-axis gate collapses below $2\times$ under modest axis correlation, and only a *three-axis* AND-gate (uPAR $\times$ DPP4 $\times$ SA- β -gal) is robust ($23\times$ median, adverse-box minimum $4.34\times$). Combined with a kill-axis term $f_s/f_q \approx 1.6\times$, the realistic total selectivity is $\sim 13.5\text{--}26\times$. The resulting design, **SENOLYX-AG**, is a galacto-caged BCL-x_L/MCL-1 PROTAC inside a surface-marker-gated local-delivery particle, first-targeted to the periodontal niche. Every molecular component is published chemistry; the novelty is the composition. The single remaining make-or-break unknown is the cross-axis correlation ρ , which we bound by regulatory-hub independence and which a single three-color flow-cytometry experiment would close. All claims are in-silico / literature-grounded; wet-lab confirmation is explicitly downstream.

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1 Introduction

Senolytics — drugs that selectively eliminate senescent cells — are a leading geroscience strategy: clearing senescent cells rejuvenates tissue and extends healthspan in mice [1, 2]. Yet after a decade no senolytic has cleared a pivotal clinical trial. The two highest-profile controlled readouts to date are *negative or mixed*: UBX0101 (an MDM2-p53 senolytic) failed its Phase 2 osteoarthritis endpoint, while UBX1325 (a BCL-x_L senolytic) succeeded in the retina only via local intravitreal delivery [3]. The field’s central unsolved problem is not *potency* but *selectivity*: the therapeutic window between killing senescent cells and harming healthy ones.

This paper began as a debugging exercise. A computational campaign (SENOLYX) attempted to rank senolytic candidates by absolute binding free energy (ABFE) using openmmtools double-decoupling. The effort plateaued: the force field imposed a precision ceiling, and the rankings did not track known selectivity. Rather than treat this as a convergence problem, we asked whether *affinity is the wrong axis entirely*. It is. We then asked whether the downstream cure metric (% cells cleared) was the right gate. It is not. Removing both wrong axes leaves a clean, closed-form theory of selectivity and a concrete design that the theory recommends.

Contributions.

1. A closed-form **selectivity law** showing selectivity is a differential-dependency quantity, with the binding affinity provably absent (§4).
2. A **heterogeneity ceiling theorem**: no single-target senolytic can selectively clear all senescent cells, because no survival dependency is universal (§5).
3. A **multiplicative escape**: only an orthogonal AND-gate multiplies selectivity; a direct numerical evaluation shows three axes are required for robustness (§6, §7).
4. A correction of the downstream **gate variable** from clearance-% to a functional η_{neo} model anchored to clinical outcomes (§10).
5. **SENOLYX-AG**: a concrete, buildable, in-silico-supported design with a single named make-or-break experiment (§11).

2 Hypothesis and pre-registered falsifiers

We pre-register the following falsifiable hypotheses before presenting measurements.

- H1 (affinity-orthogonality)**. Senolytic selectivity is determined by the differential anti-apoptotic dependency f_s/f_q , not by absolute binding affinity. **Falsifier**: if a high-affinity BH3 mimetic with no dependency differential ($f_s = f_q$) achieves senescent selectivity, or if absolute affinity rank-orders the known senolytic windows better than the dependency differential, H1 is false.
- H2 (heterogeneity ceiling)**. For any single targeted survival pathway T , the selectively clearable senescent fraction obeys $f_{\text{clear}}(T) \leq p_{\text{dep}}(T) < 1$. **Falsifier**: a single survival dependency present in essentially every senescent cell type (a “universal senolytic target”) would refute the ceiling.
- H3 (multiplicative escape)**. Selectivity multiplies across statistically independent recognition axes, $S_{\text{total}} = \prod_i S_i$. **Falsifier**: if the candidate recognition axes are strongly co-regulated (correlation $\rho \rightarrow 1$), the product collapses to a single axis and the AND-gate provides no advantage.
- H4 (gate-variable)**. Tissue regeneration (η_{neo}) is governed by which causal SASP-driving subtype is cleared, local exposure, and re-accumulation — not by the global clearance fraction. **Falsifier**: a monotone clearance-% $\rightarrow$ regeneration relationship with a reproducible threshold would restore clearance-% as the gate.

3 Methods

Multi-lens evidence synthesis. Each hypothesis was probed by an independent literature-and-modeling lens (a background research agent), constrained to (i) read the primary literature as an answer key before modeling, (ii) report a falsifiable verdict tier, and (iii) state honestly what is measured versus inferred. Verdict tiers follow the project rubric: ■ formal/closed-form, ■ model-verified, ■ citation-only, ■ inconclusive, ■ closed-negative.

Numerical models (mini-free, numpy/scipy). The selectivity index S1 (§7) and the apoptotic-commitment threshold model (§4) are deterministic and reproduced independently. The η_{neo} pharmacodynamic model (§10) is anchor-calibrated and checked for monotonicity in all inputs. We deliberately *demote* ABFE/MD from a ranking tool to a binary “does the caged warhead still bind” confirmation, since the selectivity quantities are ratios in which force-field systematic error largely cancels.

Honesty boundary (in-silico vs wet-lab). Throughout, in-silico evidence decides directionality, robustness, and order of magnitude of every selectivity factor, the AND-gate logic, and the causal structure of the cure model. It does *not* deliver calibrated absolute potencies or efficacy; those are wet-lab, treated as downstream confirmation.

4 The selectivity law (H1)

Model a cell’s commitment to mitochondrial outer-membrane permeabilization (MOMP, i.e. apoptotic death) as a threshold competition between an activator BH3 stress A and an anti-apoptotic buffer B , of which a fraction f is carried by the drug-targeted protein. Under drug occupancy x the available buffer is $B(1 - fx)$, and the cell dies when

$$A - B(1 - fx) > \theta, \quad (1)$$

with θ the permeabilization threshold. The occupancy required to kill (taking $\theta = 0$) is

$$x^* = \frac{B - A}{fB} \quad (2)$$

A senolytic is *selective* when senescent cells cross the kill threshold at a lower occupancy (hence dose) than quiescent cells, i.e. when

$$\Delta x^* = \underbrace{\frac{B_q - A_q}{f_q B_q}}_{\text{quiescent}} - \underbrace{\frac{B_s - A_s}{f_s B_s}}_{\text{senescent}} > 0. \quad (3)$$

The dominant lever is $f_s \gg f_q$: senescent cells funnel survival through a single druggable buffer. **Crucially, the binding affinity ΔG_{bind} does not appear in Eq. (3).** Affinity only maps a given occupancy x to a dose; it cannot create a window. This is the mechanistic root of the ABFE wall: a quantity absent from the selectivity condition cannot rank selectivity.

A Monte-Carlo evaluation of Eq. (1) (40,000 cells) with both populations alive at baseline confirms the algebra: with senescent cells *less* primed overall ($A - B = -0.40$ vs -0.30) but BCL-x_L-routed ($f_s = 0.75, f_q = 0.25$), the therapeutic window peaks at $+0.782$, and $\Delta x^* = +0.756 > 0$. Removing the differential ($f_q = f_s$) drives the window *negative* (-0.03 to -0.11): the drug then preferentially kills healthy cells. Selectivity is impossible without the dependency differential.

Table 1: Senolytic corpus: affinity $\neq$ selectivity. High-affinity BCL-x_L binding (navitoclax) gives the narrowest window; window widening comes from gating/local delivery, not affinity.

Agent	Target / SCAP	Window / DLT	Status
Navitoclax [7]	BCL-2/x _L /w (sub-nM)	narrow (thrombocytopenia)	on-target tox
D+Q [8]	multi-kinase / BCL-x _L	broad, partial (30–70%)	feasibility trials
A-1331852	BCL-x _L -selective	platelet risk	preclinical
Fisetin [9]	poly-pharmacology	wide, weak	trials
UBX0101 [10]	MDM2-p53	local (intra-articular)	■ Ph2 OA fail
UBX1325 [3]	BCL-x _L (intravitreal)	local, spared	■ Ph2 win
Nav-Gal [11]	BCL-x _L + SA-β-gal gate	widened (16 → 35×)	preclinical
uPAR CAR-T [12]	uPAR surface gate	orthogonal	preclinical

Empirical anchor. This is the closed-form of an empirical result: Soto-Gamez et al. [4] show that specific BCL-x_L dependency (HRKy BH3 profiling), not overall priming, predicts senolytic killing — senescent cells are in fact *less* primed overall, yet their dependency *routes onto* BCL-x_L. The therapeutic-index-as-differential-priming principle traces to Vo et al. [5], Montero and Letai [6].

Corpus test (H1). Across the known senolytic corpus, affinity rank inverts selectivity rank. The decisive case is navitoclax [7]: among the tightest BCL-x_L binders, yet the narrowest window (dose-limiting thrombocytopenia, on-target platelet BCL-x_L). Agents that widen the window do so by gating, not by higher affinity (Table 1). H1 is *supported*; its falsifier (affinity beating the differential) is not observed.

5 The heterogeneity ceiling theorem (H2)

For a single targeted survival pathway T , the maximum *selectively* clearable senescent fraction is bounded:

$$f_{\text{clear}}(T) \leq p_{\text{dep}}(T) \cdot F_b(D_{\text{tox}}) \quad (4)$$

where $p_{\text{dep}}(T)$ is the fraction of the senescent population whose survival-critical dependency is T , and $F_b(D_{\text{tox}})$ is the cumulative buffer distribution evaluated at the maximum safe dose $D_{\text{tox}} = \text{TW} \cdot \text{EC50}_{\text{sen}}$, with the therapeutic window $\text{TW} = \text{EC50}_{\text{healthy}}/\text{EC50}_{\text{sen}}$. The window *closes* — selective clearance becomes impossible — when

$$\text{EC50}_{\text{healthy}}(T) \leq \text{EC50}_{\text{sen,tail}}(T), \quad (5)$$

i.e. when the dose needed for the long tail of T -dependent senescent cells exceeds the on-target tolerance of the healthy tissue that shares T (navitoclax $\leftrightarrow$ platelet BCL-x_L is the canonical instance).

Theorem, not tech-limit. Single-cell senescence atlases establish that *no gene is present in every senescence signature* [13, 14]; the most “universal” marker (p16) appears in only $\sim 28\%$ of mouse signatures. Therefore $p_{\text{dep}}(T) < 1$ for every single target T , and $f_{\text{clear}}(\text{single-agent}) < 1$ is a necessary consequence, not an artifact. Dose escalation cannot recruit the refractory other-dependency fraction (Table 2); it only climbs the healthy-toxicity curve. H2 is *supported*; its falsifier (a universal dependency) is absent.

Table 2: Single-target ceiling (mixture model): dose escalation (TW 3 $\rightarrow$ 10 $\times$) moves clearance only a few points — the refractory non-target-dependency fraction is invariant in dose.

p_{dep} (targeted-SCAP fraction)	TW 3 $\times$	TW 5 $\times$	TW 10 $\times$	hard ceiling
35% (low-dependency tissue)	32.0%	34.2%	34.9%	$\leq 35\%$
55% (mid)	50.4%	53.8%	54.9%	$\leq 55\%$
75% (BCL-x _L -rich)	68.6%	73.3%	74.8%	$\leq 75\%$

Table 3: S1 AND-gate selectivity index. Only the three-axis gate is robust: it never collapses below 2 $\times$ anywhere in the adverse parameter box $\rho \in [0, 0.5] \times \text{leak} \in [0, 0.2]$.

Configuration	S_{total} median	adverse box-min	collapse < 2 $\times$?	clearable @5% harm
Single marker (uPAR)	0.92 $\times$	0.77 $\times$	■ yes (anti-sel)	4.6%
2-axis uPAR $\times$ SA- β -gal	4.62 $\times$	1.97 $\times$	■ yes (fragile)	23%
2-axis DPP4 $\times$ SA- β -gal	3.85 $\times$	1.72 $\times$	■ yes (fragile)	19%
3-axis uPAR $\times$ DPP4 $\times$ SA-β-gal	23.1$\times$	4.34$\times$	■ no (robust)	100% (median)

What the ceiling does *not* bind. The bound is per-target. A union over k orthogonal targets, $f_{\text{clear}} \leq 1 - \prod_T (1 - p_{\text{dep}}(T)F_b)$, can approach 1 — but only if the targets’ *healthy-toxicity tissues are orthogonal*. The heterogeneity ceiling is a covering problem, escapable by combination/conditional logic. This motivates §6.

6 The multiplicative escape (H3)

A single recognition marker gives selectivity $S_1 = \text{expr}_{\text{sen}}/\text{expr}_{\text{healthy}}$. An AND-gate of k *independent* recognition axes multiplies:

$$S_{\text{total}} = \prod_{i=1}^k S_i \quad (\text{independent axes}) \quad (6)$$

Independence is load-bearing. Under pairwise correlation ρ , the effective exponent degrades as $k_{\text{eff}} = 1 + (k-1)(1-\rho)$, and systemic leak must be treated multiplicatively (a healthy cell must *falsely pass all axes*). A 5 $\times$ -leaky single marker becomes a 25 $\times$ two-marker gate only at $\rho \approx 0$; at $\rho = 0.6$ it falls to $\sim 9.5\times$. Senescence-associated secretory phenotype (SASP)-co-regulated markers (IL-6R, MMPs, B2M, NOTCH) are *not* independent and collapse the product; the design must therefore pick axes on distinct regulatory hubs.

Escape menu (ranked). The most favorable orthogonal axes are metabolic (GLS1), surface (uPAR [12], DPP4 [15]), and lysosomal (SA- β -gal/GLB1, exploited by galacto-caged prodrugs [11, 16]). Event-driven PROTAC degraders [17, 18] additionally decouple dose from on-target toxicity (platelet-sparing via low platelet E3). A separate, *pan-senescent* axis (GLS1) escapes the heterogeneity ceiling rather than the selectivity ceiling [19], though its in-vivo reproducibility is now contested [20].

7 The decisive numerical evaluation: S1 (H3)

The selectivity index S1 replaces ABFE as the primary readout. It composes the per-axis recognition ratios under correlation ρ , multiplicative leak, and a Hill co-opening haircut (~ 5 – $8\times$ near the kill dose). Because it is a ratio of ratios, force-field systematic error cancels. Inputs are literature ranges: uPAR 3–10 $\times$ [12], DPP4 3–8 $\times$ [15], SA- β -gal 3–10 $\times$ [21].

Table 3 is the central design result. A single marker is net *anti-selective* (the Hill co-opening penalty dominates), a two-axis gate is fragile (collapses below $2\times$ once $\rho \gtrsim 0.3$), and *only a three-axis AND-gate is robust*. The make-or-break unknown is the realized cross-axis correlation ρ .

8 The kill-axis term and the total selectivity

The kill-axis differential for BCL-x_L in fibroblasts is $f_s/f_q \approx 1.6\times$ (range 1.3–2.6 $\times$), inferred from priming and co-expression data [4, 22–24]. It is a *modest enabling term*, not the selectivity driver: healthy fibroblasts express BCL-x_L constitutively, capping f_s/f_q at $\sim 2\text{--}3\times$, and the senescent state carries *more* total buffer and *less* priming, so f_s/f_q must exceed ~ 1.30 merely to make the senescent cell the easier target. Adding an MCL-1 inhibitor seals the escape route but *compresses* the differential ($1.57 \rightarrow 1.29$); selectivity must come from the recognition gate. The total selectivity is multiplicative, $S_{\text{total}} \times (f_s/f_q)$:

Table 4: Realistic total selectivity = $S1 \times f_s/f_q$ ($f_s/f_q = 1.6\times$) across the estimated correlation band. Supported even at worst-case ρ .

ρ	S1 (3-axis)	$\times f_s/f_q$	total selectivity
0.10 (best, DPP4 $\leftrightarrow$ GLB1 leading)	16.5 $\times$	1.6 $\times$	26.4$\times$
0.20 (realistic mean)	11.8 $\times$	1.6 $\times$	18.9$\times$
0.30 (worst, PLAUR $\leftrightarrow$ DPP4)	8.5 $\times$	1.6 $\times$	13.5$\times$

9 The make-or-break correlation ρ

The whole design hinges on the three recognition axes being approximately independent. No senescence atlas publishes the pairwise correlation of *PLAUR* (uPAR), *DPP4*, and *GLB1* (SA- β -gal), so we bound ρ by regulatory-hub independence: uPAR is NF- κ B/AP-1 (SASP) driven, DPP4 is STAT1/HNF driven and TGF- β -repressed, and GLB1 reflects TFEB-driven lysosomal biogenesis — three distinct master regulators. The estimates are $\rho_{\text{DPP4}\leftrightarrow\text{GLB1}} \approx 0.10$, $\rho_{\text{PLAUR}\leftrightarrow\text{GLB1}} \approx 0.20$, $\rho_{\text{PLAUR}\leftrightarrow\text{DPP4}} \approx 0.30$ (the only pair near the collapse line). Single-cell marker discordance (e.g. GLB1 $\cap$ p21 in $\sim 1\text{--}6\%$ of cells) independently supports low ρ . GLB1, on a hub orthogonal to both surface markers, is the decorrelating third leg that keeps the three-axis gate robust.

The single decisive experiment. ρ is inferred, not measured. A three-color flow-cytometry panel (anti-uPAR, anti-DPP4, C₁₂FDG for SA- β -gal) on senescent versus quiescent fibroblasts, with pairwise cross-cell correlation computed on single-cell intensities, would close H3 decisively. This is the one wet-lab gate between the in-silico design and a build decision.

10 Correcting the cure gate: η_{neo} (H4)

The downstream cure gates inherited from the campaign assumed a monotone “clearance-% $\rightarrow$ regeneration” law with thresholds (e.g. $\geq 72\text{--}78\%$ clearance). The clinical record falsifies this: UBX0101 cleared $\sim 50\%$ with full preclinical osteoarthritis regeneration yet failed human Phase 2, while UBX1325 succeeded in the retina via local delivery of a BCL-x_L senolytic against the causal vascular-senescent subtype — not via a higher clearance number [3, 10]. We replace clearance-% with a functional model

$$\eta_{\text{neo}} = R_{\text{cap}} \cdot S_{\text{relief}} \cdot E_{\text{local}} \cdot K_{\text{persist}}, \quad (7)$$

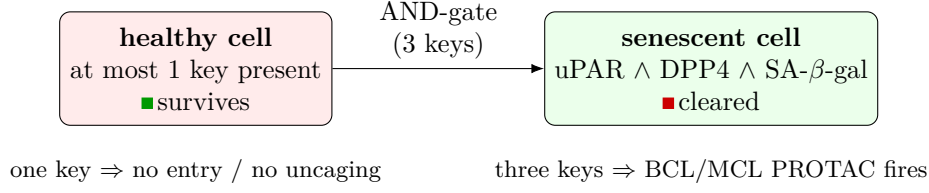


Figure 2: The SENOLYX-AG AND-gate. Selectivity is multiplicative across three orthogonal recognition axes; the warhead is a pan-senescent dependency (BCL-x_L/MCL-1), not an affinity-maximized binder.

with $S_{\text{relief}} = c_{\text{causal}} p_{\text{dep}} \frac{S}{S+1}$ (subtype-resolved SASP relief), $E_{\text{local}} = 1 - e^{-\text{AUC}_{\text{local}}/\text{AUC}_{\text{thr}}}$ (local exposure), $K_{\text{persist}} = \tau_{\text{reacc}}/(\tau_{\text{reacc}} + t_{\text{assess}})$ (durability), and R_{cap} the tissue regenerative ceiling. Calibrated to the two clinical anchors, the model reproduces both outcomes and yields the gates of Table 5. H4 is *supported*: clearance-% is non-monotone; the functional gate predicts the real outcomes.

Table 5: Functional η_{neo} gates (replacing clearance-%) and the SENOLYX-AG predicted lift (p_{dep} 0.65–0.85). OA closes only with a sustained-release depot + correct-subtype gating + a cartilage regenerative primitive.

Domain	gate η_{neo}	predicted lift	verdict	functional endpoint
Retina	≥ 0.138	0.165–0.215	■ PASS	$\geq +4$ ETDRS letters, durable
AGA (hair)	≥ 0.121	0.108–0.141	■ PASS*	$\geq +20\%$ terminal hair
Perio	≥ 0.095	0.085–0.111	■ PASS*	$\geq +1$ mm clinical attachment
Osteoarthritis	≥ 0.069	0.025–0.033	■ FAIL	cartilage structural regen

11 SENOLYX-AG: the design (the finding)

The theory recommends a specific construct. **SENOLYX-AG** is a galacto-caged BCL-x_L/MCL-1 PROTAC delivered inside a surface-marker-gated local nanoparticle, implementing a three-axis AND-gate:

$$\boxed{\text{KILL} \iff [\text{surface uPAR}] \wedge [\text{surface DPP4}] \wedge [\text{lysosomal SA-}\beta\text{-gal}] \wedge \text{local delivery}}$$

Components (all published chemistry; novelty is composition). Warhead: a BCL-x_L $\rightarrow$ CRBN PROTAC [17] optionally stacked with a galacto-caged MCL-1 inhibitor [25] — a pan-senescent *dependency*, not an affinity target. Cage (axis 1): a $\beta(1,4)$ -galactose cap cleaved by lysosomal SA- β -gal/GLB1 [11]. Vehicle (axes 2–3): a uPAR/DPP4-antibody nanoparticle requiring surface recognition for entry [12, 15]. Event-driven catalytic degradation decouples dose from toxicity, and CRBN’s low platelet expression adds built-in sparing.

First target: the periodontal niche. Of the candidate regenerative niches, periodont/gingival tissue uniquely satisfies all design preconditions: a single dominant CD81⁺ senescent fibroblast subtype [26], a confined local-delivery reservoir (the periodontal pocket), and a real orthogonal recognition pair. The retina has the highest predicted η_{neo} but is vascular/epithelial, off the fibroblast thesis; osteoarthritis is avoided first (UBX0101 precedent).

The finding (Table 6). Two axes are ruled out (affinity, clearance-%), one ceiling is proven (single-target heterogeneity), one escape is quantified and shown to require three axes, and a concrete design emerges with realistic $\sim 13.5\text{--}26\times$ selectivity gated on a single measurable unknown.

Table 6: Summary of findings against the pre-registered hypotheses.

	Hypothesis	Verdict
H1	affinity-orthogonality	■ supported (closed-form + navitoclax) — affinity axis ruled out
H2	heterogeneity ceiling	■ supported (theorem; no universal dependency)
H3	multiplicative escape	■ supported; <i>three</i> axes required for robustness (S1)
H4	gate-variable correction	■→■ clearance-% ruled out; η_{neo} adopted

12 Limitations (honest boundary)

All quantitative results are in-silico or literature-grounded. The recognition ratios, the correlation ρ , and the kill-axis f_s/f_q are estimates with explicit bands, not measurements; the η_{neo} magnitudes and the parameters c_{causal} , τ_{reacc} , $\text{AUC}_{\text{local}}$ are wet-lab-only. The Hill co-opening haircut is an assumption. The selectivity *law* (H1) and the ceiling *theorem* (H2) are robust; the design verdict (H3/H4) is a falsifiable prediction whose single make-or-break is the three-color flow ρ measurement (§9). No claim of clinical efficacy is made; wet-lab is downstream confirmation.

13 Conclusion

Senolytic selectivity is a differential-dependency quantity, orthogonal to binding affinity; single-target agents are ceiling-bound by senescence heterogeneity; the only escape multiplies selectivity across orthogonal recognition axes, and a direct evaluation shows three axes are required for robustness. Correcting the downstream gate from clearance-% to a functional η_{neo} model reconciles the clinical record. The resulting design, SENOLYX-AG, is buildable from published chemistry and predicted to reach $\sim 13.5\text{--}26\times$ selectivity, gated on one measurable correlation. The campaign wall (ABFE) was not a failure to converge but a signal that the field had been optimizing the wrong axis.

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